

NAME: BALOGUN GBENGA ISAAC

MATRIC NO: 220404222

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

IRO (Issues Reporting Outlet)

Course:

CSC426 – Software Engineering

Project Title:

IRO (Issues Reporting Outlet): A Web-Based System for Reporting

Issues Affecting AAUA Students and Host Communities

Table of Contents

Introduction

Overall Description

Functional Requirements

Non-Functional Requirements

System Architecture

Database Design

Major System Modules

Benefits of the System

Limitations

Conclusion

1.0 Introduction

1.1 Purpose

The Issues Reporting Outlet (IRO) is a web-based platform designed to enable AAUA students and members of the surrounding host communities (Akungba, Iwaro, Etioro, Ayegunle, etc.) to report issues affecting their environment. The system provides an easy, transparent, and centralized means of reporting complaints, monitoring their progress, and receiving updates from administrators.

This Software Requirements Specification (SRS) defines the functional and non-functional requirements of the proposed system and serves as a guide for its development and maintenance.

1.2 Scope

The IRO system provides an online platform for reporting and managing issues affecting both the university and host communities.

The system allows users to:

1. Register an account.

2. Log into the system.
3. Submit reports.
4. Track submitted reports.
5. View report status.

The administrator can:

- View all reports.
- Update report status.
- Manage users.
- Delete false or duplicate reports.
- Monitor activities within the system.

1.3 Objectives

The objectives of the proposed system are to:

- Improve issue reporting.
- Reduce delays associated with manual reporting.
- Encourage community participation.
- Improve transparency.
- Maintain centralized records of reported issues.
- Improve communication between students, indigenes, and administrators.

2.0 Overall Description

2.1 Product Perspective

IRO is a web-based application that can be accessed through modern web browsers on computers and smartphones.

The system consists of three major components:

- *Presentation Layer (User Interface)
- *Application Layer (Business Logic)
- *Database Layer

2.2 Product Functions

The system shall provide:

1. User Registration
2. User Login
3. Issue Reporting
4. Report Tracking
5. Administrator Dashboard
6. Status Updates
7. Report Management

2.3 User Classes

Student

Students can:

- Register
- Login
- Submit reports
- Track reports
- View report history

Community Indigene

Community members can:

- Register
- Login
- Submit reports
- Track reports
- View report history

Administrator

The administrator can:

- Login securely
- View all reports
- Update report status
- Delete reports
- Manage users
- Generate report summaries

2.4 Operating Environment

The system supports:

- Windows
- Linux
- macOS
- Android
- iOS

Compatible browsers include:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge
- Safari

2.5 Assumptions

The system assumes that:

- Users have internet access.
- Users possess basic computer knowledge.
- The web server remains operational.
- Administrators regularly review submitted reports.

3.0 Functional Requirements

FR1 User Registration

The system shall allow users to register using:

- Full Name
- Email Address
- Phone Number
- Password
- User Type (Student or Community Indigene)

FR2 User Login

The system shall authenticate registered users before granting access.

Inputs:

- Email

- Password

Outputs:

- Successful Login

- Invalid Login Message

FR3 Submit Issue

Authenticated users shall submit reports containing:

- Issue Title

- Category

- Description

- Location

- Date

- Optional Image Upload

FR4 View Submitted Reports

Users shall be able to:

- View submitted reports.

- Check current report status.

- Read administrator feedback.

FR5 Track Issue

The system shall display report status such as:

- Pending

- Under Review

- In Progress

- Resolved

FR6 Administrator Login

- Only authorized administrators shall access the administration dashboard.

FR7 Manage Reports

Administrators shall:

- Approve reports.

Reject false reports.

- Assign priorities.

- Update report status.

- Delete duplicate reports.

FR8 Search Reports

Administrators shall search reports using:

- Category

- Date

- Status

- Reporter Name

- Community

4.0 Non-Functional Requirements

4.1 Performance

- *Pages should load within three seconds under normal conditions.

- *The system should support multiple simultaneous users.

4.2 Security

- *Passwords should be encrypted.
- *Authentication is required before access.
- *Unauthorized users cannot access the administrator dashboard.

4.3 Reliability

- *The system should operate with minimal downtime and maintain data integrity.

4.4 Availability

- *The application should be available 24 hours a day except during scheduled maintenance.

4.5 Usability

- *The interface should be simple, intuitive, and user-friendly for both students and community members.

4.6 Maintainability

- *The application should be easy to maintain, update, and expand in the future.

4.7 Portability

- *The application should function correctly across different operating systems and modern web browsers.

5.0 System Architecture

The architecture consists of three layers.

1.Presentation Layer

2.Home Page

3.Login Page

Report Issue Page

5.0 SYSTEM ARCHITECTURE

The system architecture of the IRO (Issues Reporting Outlet) is designed using a three-tier architecture comprising the Presentation Layer, Application Layer, and Database Layer.

Layer	Components	Description	Technologies
 Presentation Layer (User Interface)	• Home Page • User Registration Page • User Login Page • Report Issue Page • User Dashboard • Administrator Dashboard	This layer handles all user interactions. It provides web pages and interfaces that allow users (students and community members) to report issues, track reports, and view updates, while administrators manage reports and users.	• HTML5 • CSS3 • JavaScript • Bootstrap 5
 Application Layer (Business Logic)	• Authentication Module • Report Management Module • User Management Module • Status Management Module • Notification Module • Search & Filtering Module	This layer contains the business logic and rules that process user requests, validate data, manage reports, handle user accounts, and update report statuses.	• PHP 8 / JavaScript • Node.js (Optional) • AJAX • JSON
 Database Layer (Data Storage)	• Users Table • Reports Table • Categories Table • Status Table • Feedback Table • Administrator Table	This layer stores all system data including users, reports, categories, statuses, feedback, and administrator information in a secure database.	• MySQL • phpMyAdmin

6.0 Database Design

6.0 DATABASE DESIGN

The system database is designed to store and manage all data related to users, reports, categories, and administrators. The main tables and their attributes are described below.

6.1 Users Table

Field Name	Data Type	Size	Constraints	Description
UserID	INT	11	Primary Key, Auto Increment	Unique ID for each user
FullName	VARCHAR	100	NOT NULL	Full name of the user
Email	VARCHAR	100	NOT NULL, UNIQUE	Email address of the user
Phone	VARCHAR	15	NOT NULL	Phone number of the user
Password	VARCHAR	255	NOT NULL	Encrypted password
UserType	ENUM	-	NOT NULL	Student or Community Indigene
DateRegistered	DATETIME	-	NOT NULL, Default: CURRENT_TIMESTAMP	Date and time of registration

6.2 Reports Table

Field Name	Data Type	Size	Constraints	Description
ReportID	INT	11	Primary Key, Auto Increment	Unique ID for each report
UserID	INT	11	Foreign Key (Users.UserID)	ID of the user who reported
Title	VARCHAR	150	NOT NULL	Title of the issue
CategoryID	INT	11	Foreign Key (Categories.CategoryID)	Category of the issue
Description	TEXT	-	NOT NULL	Detailed description of the issue
Location	VARCHAR	150	NOT NULL	Location of the issue
Status	ENUM	-	NOT NULL, Default: 'Pending'	Status of the report
DateReported	DATETIME	-	NOT NULL, Default: CURRENT_TIMESTAMP	Date and time the report was submitted
ImagePath	VARCHAR	255	NULL	Path to uploaded image (if any)
AdminRemark	TEXT	-	NULL	Feedback from administrator
LastUpdated	DATETIME	-	NOT NULL, Default: CURRENT_TIMESTAMP	Last time the report was updated

6.3 Categories Table

Field Name	Data Type	Size	Constraints	Description
CategoryID	INT	11	Primary Key, Auto Increment	Unique ID for each category
CategoryName	VARCHAR	100	NOT NULL, UNIQUE	Name of the category
Description	VARCHAR	255	NULL	Description of the category

6.5 Feedback Table

Field Name	Data Type	Size	Constraints	Description
FeedbackID	INT	11	Primary Key, Auto Increment	Unique ID for each feedback
ReportID	INT	11	Foreign Key (Reports.ReportID)	ID of the related report
AdminID	INT	11	Foreign Key (Administrator.AdminID)	ID of the admin who gave feedback
Feedback	TEXT	-	NOT NULL	Feedback or comment
DatePosted	DATETIME	-	NOT NULL, Default: CURRENT_TIMESTAMP	Date and time feedback was posted

6.4 Status Table

Field Name	Data Type	Size	Constraints	Description
StatusID	INT	11	Primary Key, Auto Increment	Unique ID for each status
StatusName	VARCHAR	50	NOT NULL, UNIQUE	Name of the status
Description	VARCHAR	255	NULL	Description of the status

6.6 Administrator Table

Field Name	Data Type	Size	Constraints	Description
AdminID	INT	11	Primary Key, Auto Increment	Unique ID for each administrator
Username	VARCHAR	50	NOT NULL, UNIQUE	Administrator username
Password	VARCHAR	255	NOT NULL	Encrypted password
FullName	VARCHAR	100	NOT NULL	Full name of the administrator
Email	VARCHAR	100	NOT NULL, UNIQUE	Email address of administrator
Phone	VARCHAR	15	NULL	Phone number of administrator
DateCreated	DATETIME	-	NOT NULL, Default: CURRENT_TIMESTAMP	Date and time account was created

7.0 Major System Modules

The system is divided into the following modules:

- 1.Registration Module
- 2.Login Module
- 3.Issue Reporting Module
- 4.Report Tracking Module
- 5.Administrator Module
- 6.Database Management Module

8.0 Benefits of the System

The proposed system provides the following benefits:

- 1.Faster issue reporting.
- 2.Better communication between students, community members, and administrators.
- 3.Centralized storage of reports.
- 4.Reduced paperwork.
- 5.Improved transparency and accountability.
- 6.Easier monitoring of reported issues.

9.0 Limitations

The system has the following limitations:

1. Internet access is required.
2. The quality of service depends on administrator responsiveness.
3. The system facilitates reporting but does not resolve issues automatically.

10.0 Conclusion

The IRO (Issues Reporting Outlet) system provides a reliable, user-friendly, and efficient platform for reporting and monitoring issues affecting AAUA students and surrounding host communities. By digitizing the reporting process, the system improves communication, enhances accountability, and promotes timely responses to reported concerns. The proposed solution supports better record management and contributes to a more responsive and transparent reporting process.